CHAPTER 6

ELECTROMAGNETIC INDUCTION

CHAPTER ESSENCE AT A GLANCE:
Faraday's discovery of induced electromotive force from changing magnetic flux, Lenz's law, motional EMF, and self and mutual inductance.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Faraday's Law of Induction:** A time-varying magnetic flux linked with a conducting circuit induces an electromotive force proportional to the time rate of change of magnetic flux.

• **Lenz's Law & Energy Conservation:** Induced current flows in such a direction that its magnetic effect opposes the change in magnetic flux that produced it, ensuring strict conservation of energy.

• **Motional Electromotive Force:** Moving a conducting rod across a perpendicular magnetic field exerts magnetic Lorentz forces on internal free charges, driving them to establish an induced voltage.

• **Self and Mutual Inductance:** Inductance represents electrical inertia opposing current changes; self-inductance opposes variations in the same coil, while mutual inductance couples current into a secondary coil.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\epsilon = -\frac{d\Phi_B}{dt} = -N\frac{d\Phi_B}{dt}$$

Faraday-Lenz Law of electromagnetic induction

$$\epsilon = Bvl$$

Motional electromotive force (EMF) across a moving conductor

$$e = -L\frac{dI}{dt} \quad \text{and} \quad U = \frac{1}{2}LI^2$$

Self-induction EMF and magnetic energy stored in an inductor

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Lenz's law explains that mechanical work done against opposing magnetic force transforms into electrical energy.

• **Alert:** [Valuation Rubric] Step 1 [Diagram & Setup]: Draw conductor of length l moving on U-shaped rails in uniform perpendicular field B .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Faraday's laws of electromagnetic induction [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Faraday's Laws of Electromagnetic Induction]** 1. First Law: Whenever magnetic flux linked with a closed conducting loop changes with time, an electromotive force (EMF) is induced in the loop. The induced EMF persists as long as the change in flux continues. [1 Mark]

• 2. Second Law: The magnitude of the induced EMF is directly proportional to the time rate of change of magnetic flux linked with the circuit: e = - dΦ B divided by dt. For N turns: e = - N (dΦ B divided by dt). Negative sign represents Lenz's Law (direction of induced EMF opposes the flux change). [1 Mark]

• **Key Exam Result:** e = - dΦ B divided by dt; For N turns: e = - N (dΦ B divided by dt)

2. Lenz's law and its justification using conservation of energy [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Lenz's Law Statement]** The polarity of the induced EMF is such that it tends to produce an induced current whose magnetic field opposes the change in magnetic flux that produces it. [1 Mark]

• **[Energy Conservation Proof]** When the North pole of a bar magnet is pushed towards a coil, induced current flows counter-clockwise, producing a North pole on the near face. This repulsive force opposes the inward motion of the magnet. Mechanical work done by external agent against this magnetic repulsion is converted into electrical energy (induced current), which dissipates as Joule heat (I^2Rt). If induced current attracted the magnet instead, the magnet would accelerate without any external work, creating energy from nothing, violating the First Law of Thermodynamics! Hence Lenz's law is a consequence of Conservation of Energy. [1 Mark]

• **Key Exam Result:** Lenz's Law obeys Conservation of Energy (mechanical work --> electrical energy)

3. Motional EMF in a conductor moving in magnetic field e = Blv [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup & Derivation]** Straight conducting rod PQ of length L moving with constant velocity v perpendicular to uniform magnetic field B directed into page along rails. [1 Mark]

• **[Method 1: Lorentz Force Equilibrium]** Free electrons inside rod experience downward magnetic Lorentz force: F m = e v B. Electrons accumulate at end Q, creating upward electrostatic field E. In steady state: F e = F m implies e E = e v B implies E = v B. Induced EMF across ends: e = E · L = B L v. [1 Mark]

• **[Method 2: Flux Cutting]** In time dt, rod sweeps area dA = L dx = L (v dt). Flux cut: dΦ = B dA = B L v dt. Induced EMF magnitude: e = dΦ divided by dt = B L v. [1 Mark]

• **Key Exam Result:** Motional EMF e = B L v (Direction given by Fleming's Right-Hand Rule)

4. Induced EMF in a rotating rod in magnetic field e = half (0.5) B \ω l^2 [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup]** Conducting rod OA of length l pivoted at center O rotating with uniform angular velocity ω in a plane perpendicular to uniform magnetic field B. [1 Mark]

• **[Step 1: Integration over Radial Elements]** Consider small element of length dr at distance r from pivot O. Linear speed of element: v(r) = r ω. Motional EMF induced across element dr: de = B v(r) dr = B (r ω) dr. Total induced EMF between center O and tip A: e = $\int_0^l$ B ω r dr = B ω [r^2 divided by 2]$_0^l$ = (half (0.5)) B ω l^2. Since ω = 2π ν: e = (half (0.5)) B (2π ν) l^2 = B (π l^2) ν = B A ν (where A = π l^2 is swept circular area). [2 Marks]

• **Key Exam Result:** Induced EMF e = (half (0.5)) B ω l^2 = B (π l^2) ν

5. Self-inductance: definition, formula, unit, and self-inductance of a long solenoid [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Self-Inductance Definition]** Phenomenon in which an induced EMF is set up in a coil due to a change of current in the same coil: Φ = L I implies e = - L (dI divided by dt). Coefficient of self-inductance L = - e divided by (dI divided by dt). SI unit: Henry (H = V s A^{-1}). [1 Mark]

• **[Derivation for Long Solenoid]** Solenoid of length l, area A, n turns per unit length, total turns N = n l. Magnetic field inside: B = μ_0 n I = μ_0 (N divided by l) I. Flux through each turn: Φ_1 = B A = [μ_0 (N divided by l) I] A. Total flux linkage: N Φ_1 = N [μ_0 (N divided by l) I A] = [μ_0 N^2 A divided by l] I. Comparing with total flux = L I: L = μ_0 N^2 A divided by l = μ_0 n^2 A l. With core of relative permeability μ r: L = μ_0 μ r N^2 A divided by l. [2 Marks]

• **Key Exam Result:** L = μ_0 N^2 A divided by l = μ_0 n^2 A l; Increases with N^2, A, μ r and decreases with l

6. Numerical on motional EMF and induced current [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula & Given Data]** Motional EMF e = B L v. Induced current I = e divided by R = B L v divided by R. Given rod length L = 1.0 m, speed v = 5.0 m/s, B = 0.5 T perpendicular to motion, resistance R = 2.5 Ω. [1.5 Marks]

• **[Step 2: Calculations]** 1. Induced EMF: e = 0.5 × 1.0 × 5.0 = 2.5 V. 2. Induced current: I = 2.5 V divided by 2.5 Ω = 1.0 A. 3. Retarding magnetic force: F = I L B = 1.0 × 1.0 × 0.5 = 0.5 N. 4. Power dissipated: P = I^2 R = $(1.0)^2$ × 2.5 = 2.5 W (matches mechanical power F v = 0.5 × 5 = 2.5 W). [1.5 Marks]

• **Key Exam Result:** Induced EMF e = 2.5 V; Current I = 1.0 A; Power P = 2.5 W

7. Numerical on Faraday's law \Δ \ϕ divided by \Δt [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Problem & Formulas]** Coil of N = 500 turns, area A = 4.0 × 10^{-2} m^2. Initial field B_1 = 0.2 T, reversed to B_2 = -0.2 T in time Δt = 0.1 s. [1 Mark]

• **[Step 2: Substitution & Evaluation]** Initial flux per turn: Φ_1 = B_1 A = 0.2 × 4.0 × 10^{-2} = 8.0 × 10^{-3} Wb. Final flux per turn: Φ_2 = B_2 A = -0.2 × 4.0 × 10^{-2} = -8.0 × 10^{-3} Wb. Change in flux: $\Delta\Phi$ = Φ_2 - Φ_1 = -8.0 × 10^{-3} - 8.0 × 10^{-3} = -1.6 × 10^{-2} Wb. Induced EMF: |e| = N |$\Delta\Phi$ divided by Δt| = 500 × |(-1.6 × 10^{-2}) divided by 0.1| = 500 × 0.16 = 80 V. [1 Mark]

• **Key Exam Result:** Induced EMF |e| = 80 V